DISCRETE MATHEMATICS

(234202 - 234208)

1)p->q is logically equivalent to

(a)~q->~p (b)~q->p (c)p->~q (d)~p->~q

Explanation:

- **p -> q** is read as "if p, then q". This is called an *implication*.
- The statement **~q -> ~p** is the *contrapositive* of p -> q.
- The contrapositive of a conditional statement is logically equivalent to the original statement.

Therefore, p -> q is logically equivalent to ~q -> ~p.

2) p: it is below freezing

Q: it is snowing

It is snowing follow from it is below freezing

(a)p->q (b)p<->q (c)p^q (d)p XOR q

Explanation:

- **p:** It is below freezing
- **q:** It is snowing

The statement "It is snowing follows from it is below freezing" implies that if it's below freezing (p), then it will snow (q). This is the definition of implication in logic.

Therefore, the correct logical expression is **p -> q**.

3) x+y=z. is ___

(a)declarative (b)proposition (c) A and B (d)open statement

Explanation:

- **Declarative sentence:** A declarative sentence makes a statement. However, "x+y=z" doesn't make a complete statement.

- **Proposition:** A proposition is a declarative sentence that is either true or false. Since the truth value of "x+y=z" depends on the values of x, y, and z, it cannot be a proposition.
- **Open statement:** An open statement contains variables and becomes a proposition only when specific values are assigned to those variables. This accurately describes "x+y=z".

Therefore, the correct answer is (d) open statement.

4)__propositional STATEMENT_ is a declarative sentence is either true or false but not both.

5) disjunction is also known as

(a)inclusive or (b)exclusive or (c)conditional (d)biconditional

6) p v(p^q)=____

(a)p (b)q (c)p v q (d) p^q

p v (p ^ q) = p

This is a logical equivalence in propositional logic. It can be proven using truth tables or algebraic manipulation of logical expressions.

Essentially, if p is true, then the entire expression is true regardless of the value of q. If p is false, the entire expression is false. Therefore, the value of the expression depends solely on the value of p.

So, the correct answer is **(a) p**.

7) p->(q^r) is logically equivalent to

(a) (p->q)^(p-> r) (b) (p->r)^(p-> q) (c) (p->q) v (p-> r) (d) (p->r)^(p-> q)

The answer is (a) (p->q)^(p->r).

This is a classic logical equivalence known as the **distribution of implication over conjunction**.

Here's a breakdown of why:

- **p -> (q^r)** means "If p is true, then both q and r must be true."
- **(p->q)^(p->r)** means "If p is true, then q is true, and if p is true, then r is true."

8) ~(p->q) is equivalent to

(a) (~p v q)   (b) (p v q)   (c) (p v ~q)   (d) (p ^ q)

~(p->q) is equivalent to (a) (~p v q)

Here's a breakdown:

- **p -> q** is equivalent to **~p v q** (this is a fundamental equivalence in logic)
- **~(p -> q)** is the negation of the above, so it's equivalent to **~(~p v q)**
- Applying De Morgan's law to **~(~p v q)**, we get **p ^ ~q**

Therefore, **~(p -> q)** is equivalent to **p ^ ~q**.

9) s(x): x is a student of the class

C(x): x studied calculus.

One student in this class has studied calculus

[connect this in logical expression]

(a) x[s(x)-> c(x)] (b) x[s(x)-> c(x)] (c) x x [s(x)-> c(x)] (d) [c(x)-> s(x)]

Explanation:

- **s(x):** x is a student of the class
- **c(x):** x studied calculus

The statement "One student in this class has studied calculus" means that there exists at least one student in the class who studied calculus.

- **x:** There exists an x (a student)
- **s(x) -> c(x):** If x is a student, then x studied calculus.

Therefore, the correct logical expression is: **x[s(x) -> c(x)]**

This reads as "There exists an x such that if x is a student, then x studied calculus."

10) p-> q

~p is example of

~q

(a)fallacy of denying anticident

(b)modus ponens

(c)fallacy of affirming the concequence

(d) modus tollens

Explanation:

- The given argument form is:
  - p -> q
  - ~p
  - Therefore, ~q
- This is a classic example of the **fallacy of denying the antecedent**.
  - In this fallacy, one denies the antecedent (p) of a conditional statement (p -> q) and incorrectly concludes that the consequent (q) must also be false.
  - However, denying the antecedent does not necessarily lead to the negation of the consequent.

Therefore, the correct answer is (a) fallacy of denying the antecedent

11) p v q

~q is example of

:. **P**

(a)disjunctive syllogism

(b)modus tollens

(c)modus pones

(d) simplification

Explanation:

- **Disjunctive syllogism** is a valid argument form in logic where:
  - If we know that either P or Q is true (P v Q), and
  - We know that Q is false (~Q),
  - Then we can conclude that P must be true.

This matches the given argument form:

- P v Q (either P or Q is true)
- ~Q (Q is false)
- Therefore, P (P must be true)

12) x p(x)

:. P(c) for some element c is an example of

(a)universal generalization

(b)universal specification

(c)existential generalization

(d)existential specification

Explanation:

- **x p(x)** means "There exists an x such that p(x) is true." This is an existential statement.
- **P(c) for some element c** takes this existential statement and specifies a particular instance of it.

This process of going from a general existential statement to a specific instance is called **existential specification**.

Therefore, the correct answer is (d) existential specification.

13) if p is true and r is false the p XOR r = _____ (TRUE) odd no.of true

14) “ a sufficient conditional for q is p ” is example of _______(CONDITIONAL STATEMENT)

15) compound proposition p and q are logically equivalent if p->q is a ________(TAUTOLOGY)

A tautology is a statement that is always true, regardless of the truth values of its components.

In this case, p->q (p implies q) is a tautology if and only if whenever p is true, q is also true. This is the exact definition of logical equivalence between p and q.

16) In function Q(x, y) x and y are variable and q is a _____(PREDICATE)

/v (µv }v Y~˘U ˙ U ˘ v ˙ WÀ Z] /o Xd v]

17) the given notation x p(x), is known as ______{UNIVERSAL QUANTIFER}

18) propositions that cannot be expressed in terms of simpler propositions are called _____(AUTOMIC PROPOSITIONS)

19) find the negation of the proposition:

“ vandana’s smart phone has atleast 32 GB of memory ” and express this in simple english

s v v u Z}v••Zš v }(u u} ˙X

ʁɹ˝ úÙġ ē ÀĀÓ ĕ ÌÙ ġëÙ ēĖĆēĆĚíġíĆĀĚ: ¢ëÙ ÎĆĀÓíġíĆĀÀú ĚġÀġÙ ÿ ÙĀġ ē-¸ ĕ íĚ ġëÙ ēĖĆēĆĚíġíĆĀ ˧ íæ ē` ġëÙĀ ĕ ˩` ġëÙ ÎĆĀÓíġíĆĀÀú ĚġÀġÙ ÿ ÙĀġ ē-¸ ĕ íĚ æÀúĚÙ ĲëÙĀ ē íĚ ġĖĥÙ ÀĀÓ ĕ íĚ æÀúĚÙ` ÀĀÓ ġëÙ ġĖĥÙ ĆġëÙĖĲíĚÙ íĀ ġëÙ ÎĆĀÓíġíĆĀÀú ĚġÀġÙ ÿ ÙĀġ ē-¸ ĕ ` ē ˩Ě ÎÀúúÙÓ ġëÙ ------ ÀĀÓ ĕ íĚ ÎÀúúÙÓ ġëÙ -----

˜K·s f¢K8z z` - f ` -Z¦z¢ f ` ˝

p->q p(hypothesis) q(conclusion)

ʁɺ˝ ¢ëÙ ĚġÀġÙ ÿ ÙĀġ ē-¸ ĕ íĚ ÎÀúúÙÓ À ÎĆĀÓíġíĆĀÀú ĚġÀġÙ ÿ ÙĀġ ÌÙÎÀĥĚÙ ē-¸ ĕ ÀĚĚÙĖġĚ ġëÀġ ĕ íĚ ġĖĥÙ ÀĚĚÙĚĚ ÿ ÙĀġ ĆĀ ġëÙ ÎĆĀÓíġíĆĀ ġëÀġ ē ëĆúÓĚ: ÎĆĀÓíġíĆĀÀú ĚġÀġÙ ÿ ÙĀġ íĚ ÀúĚĆ ÎÀúúÙÓ ÀĀ ------

˜/u o] ˝: }v

ʁʁ˝ ġëÙ ēĖĆēĆĚíġíĆĀ ĕ-¸ ē íĚ ÎÀúúÙÓ ġëÙ ÎĆĀıÙĖĚÙ Ćæ ē-¸ ġëÙ

q->p = p->q (converse)
p->q = ~q->~p (contrapositive)
~p->~q = p->q (invertion)

ʁʂ íĚ ġëÙ ēĖĆēĆĚíġíĆĀ ˧m ē

- **Contrapositive:** The contrapositive of p -> q is **~q -> ~p**.
- **Inverse:** The inverse of p -> q is **~p -> ~q**.
- ^}U Z u v }µ Z }v }] À v]vÀ } U µ Z
  Z }vÀ]]v } X

ꞓw˝ ÓĖÀĲ ġëÙ úĆçíÎÀú ĚĴ ÿ ÌĆú æĆĖ f v çÀġÙ:

ƒ – ‡ › • „ ' Ž

or gate = +
and gate = .

Here's the standard symbol for an OR gate:

ꞓy˝ ÀÎÎĆĖÓíĀç ġĆ 2Ù _ĆĖçÀĀ úÀĲĚ`˩.. ˜ē–ĕ˝˩ íĚ ÙĕĥíıÀúÙĀġ ġĆ -----˜.. ē².. ĕ˝

}]vP } D} P v o ` U ~] µ]À o v } ~

ꞓ'˝ úÙġ ē˜ĳ˝ ÓÙĀĆġÙ ġëÙ ĚġÀġÙ ÿ ÙĀġ ˧ĳ˛_ ʸ˩: µëÀġ ÀĖÙ ġëÙĚÙ ġĖĥġë ıÀúĥÙĚ˙

- The antecedent is (p->q) ^ q
- The consequent is p